

A Comparative Quantitative Analysis of the U.S. and Kazakhstan Stock Markets

Independent Research Project

Author:
Zhanbota Erlan

National Physics and Mathematics School
Almaty, Kazakhstan

Email: fireinthehole973@gmail.com

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Risk, Return, Diversification, and Trend-Following Strategies: Evidence from U.S. Technology Stocks and the Kazakhstan Equity Market

Chapter 1: Introduction

Investment decision-making fundamentally involves balancing risk and return. Investors seek assets that can generate attractive returns while minimizing exposure to uncertainty. However, higher expected returns are often associated with greater levels of risk, making the trade-off between risk and return one of the most important concepts in finance. Understanding this relationship is essential for both individual and institutional investors when constructing investment portfolios and evaluating investment opportunities.

One of the most influential concepts in modern finance is diversification. According to Modern Portfolio Theory (MPT), developed by Harry Markowitz, investors can reduce portfolio risk by combining assets that are not perfectly correlated. Through diversification, losses from one investment may be partially offset by gains from another, resulting in improved risk-adjusted performance. As a result, diversification has become one of the most widely used risk-management techniques in financial markets. Evaluating the effectiveness of diversification across different markets remains an important area of financial research.

In addition to portfolio diversification, investors frequently employ technical trading strategies to improve investment performance. Among these strategies, moving-average-based trend-following systems are particularly popular because of their simplicity and widespread use. These strategies attempt to identify market trends by comparing current prices with historical averages, generating trading signals that may help investors reduce losses during market downturns or capture gains during upward trends. Nevertheless, the effectiveness of moving-average strategies remains uncertain and may vary across different stocks and market environments.

The United States represents one of the largest, most liquid, and most developed equity markets in the world. The market hosts globally influential technology companies such as Apple, Microsoft, and NVIDIA, which have experienced significant growth over recent years due to technological innovation, cloud computing, and the rapid expansion of artificial intelligence. Because of their size, liquidity, and global importance, these firms provide valuable examples for studying investment performance in a developed market environment.

In contrast, Kazakhstan represents an emerging market with a smaller number of listed companies and different economic characteristics. Major Kazakhstan firms such as Kaspi.kz, Halyk Bank, and Kazatomprom operate in sectors including fintech, banking, and natural resources, which play important roles in the country's economy. Compared with the United States, Kazakhstan's equity market has received considerably less academic attention, despite offering potentially valuable insights into investment behavior in emerging-market conditions.

Comparing developed and emerging markets is important because differences in market structure, liquidity, investor participation, and economic development may lead to distinct patterns of risk, return, correlation, and diversification benefits. Furthermore, trading strategies that perform effectively in mature markets may not necessarily produce similar results in emerging markets. Therefore, a comparative analysis can help investors better understand how investment opportunities and portfolio-management techniques vary across different market environments.

This study conducts a comparative quantitative analysis of selected U.S. and Kazakhstan stocks. Specifically, it examines risk-return characteristics, diversification benefits, portfolio performance, and the effectiveness of moving-average-based trend-following strategies. Historical stock price data are analyzed using measures such as average returns, volatility, correlation coefficients, portfolio risk-adjusted performance, and moving-average trading rules. By comparing the results across the two markets, this research aims to provide a deeper understanding of how investment characteristics differ between developed and emerging equity markets.

Research Questions

1. How do risk and return characteristics differ between U.S. and Kazakhstan stocks?
2. Do diversification benefits exist in both markets?
3. How does portfolio construction affect risk-adjusted performance?
4. How does moving-average length affect the performance of trend-following strategies?

Chapter 2: Literature Review

2.1 Risk and Return

Risk and return are central concepts in finance (Bodie, Kane, & Marcus, 2021). Investors generally expect higher returns as compensation for taking higher levels of risk. Return measures the gain or loss generated by an investment over a specific period, while risk reflects the uncertainty or variability of those returns. In empirical finance, return is commonly measured using percentage price changes, and risk is often represented by volatility, which is calculated as the standard deviation of returns.

The risk-return trade-off suggests that assets with greater uncertainty should offer higher expected returns. However, realized returns may differ from expected returns, especially in equity markets where prices are affected by macroeconomic conditions, firm performance, investor sentiment, interest rates, and global market events. Therefore, comparing mean returns and volatility across assets allows researchers to evaluate whether higher returns are associated with higher risk.

In this research, risk and return are analyzed using daily stock price data. Mean daily returns are used to evaluate average performance, while volatility is used to measure the degree of fluctuation in returns. The return-to-risk ratio is also used as a simple measure of risk-adjusted performance, allowing comparison between stocks with different levels of volatility.

2.2 Modern Portfolio Theory

Modern Portfolio Theory (MPT), introduced by Harry Markowitz in 1952, is one of the most important foundations of portfolio management. The theory argues that investors should not evaluate assets only individually, but should consider how assets interact within a portfolio. According to MPT, portfolio risk depends not only on the risk of each individual asset, but also on the correlations among asset returns.

A key contribution of Modern Portfolio Theory is the idea that combining assets with imperfect correlations can reduce overall portfolio risk. If two assets do not move exactly together, losses in

one asset may be offset by gains or smaller losses in another. This means that a portfolio can achieve better risk-adjusted performance than a single asset, even if the individual assets are risky.

These principles of Modern Portfolio Theory provide a theoretical foundation for evaluating portfolio construction and diversification. Equal-weighted portfolios and correlation analysis are commonly employed to assess whether combining assets can reduce volatility and risk-adjusted returns.

2.3 Diversification

Diversification is a practical application of Modern Portfolio Theory. It involves spreading investment across different assets, sectors, or markets in order to reduce unsystematic risk. Unsystematic risk is firm-specific or industry-specific risk that can be reduced through portfolio construction. In contrast, systematic risk affects the broader market and cannot be fully eliminated through diversification.

The effectiveness of diversification depends heavily on correlation. When assets have low or negative correlations, diversification benefits tend to be stronger. When assets are highly correlated, diversification benefits are weaker because the assets tend to rise and fall together. Therefore, correlation analysis is essential for understanding whether a group of stocks can provide meaningful risk reduction.

In developed markets such as the United States, large technology companies may be highly connected due to common exposure to innovation cycles, interest rates, investor sentiment, and global economic conditions. In emerging markets such as Kazakhstan, companies may operate in different sectors such as fintech, banking, and natural resources, which may create different correlation patterns. Consequently, comparing developed and emerging markets provides an opportunity to evaluate different market environments.

2.4 Trend-Following Strategies

Trend-following strategies are widely used in financial markets. These strategies are based on the idea that asset prices may continue moving in the same direction for a period of time. Instead of predicting fundamental value, trend-following strategies attempt to identify existing price trends and generate trading signals based on those trends.

One common trend-following tool is the moving average. A moving average smooths price data by calculating the average price over a selected period, such as 20, 50, 100, or 200 days. A simple moving-average strategy may generate a buy signal when the current price is above the moving average and move out of the asset when the current price is below the moving average. Shorter moving averages, such as MA20, respond more quickly to price changes, while longer moving averages, such as MA200, respond more slowly and may capture longer-term trends.

The effectiveness of moving-average strategies is debated. Brock, Lakonishok, and LeBaron in 1992 found that moving-average trading rules generated abnormal returns in U.S. equity markets, although subsequent studies have reported mixed findings regarding their effectiveness across different assets and market conditions. In some markets, they may help reduce losses during downward trends. However, they may also generate false signals, especially in sideways or highly volatile markets. Transaction costs, liquidity, and market efficiency can also affect their real-world performance. In this study, moving-average strategies are tested using different moving-average lengths to examine whether trend-following performance differs across U.S. and Kazakhstan stocks.

2.5 Developed vs Emerging Markets

Developed and emerging equity markets often differ in market size, liquidity, regulation, investor participation, information efficiency, and economic structure. Developed markets, such as the United States, generally have higher liquidity, broader investor participation, stronger disclosure standards, and more efficient price discovery. These characteristics may reduce pricing inefficiencies and influence the performance of trading strategies. According to the World Bank(2024), developed markets generally exhibit higher liquidity, stronger institutional participation, and greater market efficiency than emerging markets.

Emerging markets, such as Kazakhstan, may have lower liquidity, fewer listed firms, greater sensitivity to domestic economic conditions, and higher exposure to sector-specific risks. These features can lead to different patterns of volatility, correlation, and return behavior. Emerging markets may offer higher growth potential, but they may also involve higher uncertainty and lower market efficiency. The Astana International Financial Center(2024) also noted that Kazakhstan's financial sector continues to develop and remains relatively concentrated compared with major global markets.

Comparing U.S. and Kazakhstan equity markets is therefore useful for understanding how investment characteristics vary across different market environments. U.S. technology stocks such as Apple, Microsoft, and NVIDIA represent large, liquid firms in a developed market. Kazakhstan-related stocks such as Kaspi.kz, Halyk Bank, and Kazatomprom provide exposure to an emerging market with different economic drivers. This contrast allows the study to evaluate whether risk-return behavior, diversification benefits, and moving-average strategy performance are similar or different across market types.

Chapter Summary

This chapter reviewed the main theoretical concepts related to the study: risk and return, Modern Portfolio Theory, diversification, trend-following strategies, and the differences between developed and emerging markets. These theoretical concepts establish the analytical framework used throughout this study and provide a basis for interpreting the empirical findings presented in subsequent chapters. Based on this literature, the study examines whether selected U.S. and Kazakhstan stocks differ in terms of return, volatility, correlation, portfolio performance, and moving-average strategy effectiveness.

Reference

Bodie, Z., Kane, A., & Marcus, A. J. (2021). Investments.

Markowitz, H. (1952). Portfolio Selection. The Journal of Finance, 7(1), 77–91.

Brock, W., Lakonishok, J., & LeBaron, B. (1992). Simple Technical Trading Rules and the Stochastic Properties of Stock Returns. The Journal of Finance, 47(5), 1731–1764.

Fama, E. F. (1970). Efficient Capital Markets: A Review of Theory and Empirical Work.

World Bank. (2024). Global Financial Development Report.

Data methodology

3.1 Data Collection

This study uses historical stock price data to conduct a comparative quantitative analysis of selected U.S. and Kazakhstan equities. The data were collected from Yahoo Finance, which provides publicly available historical price information for listed companies across global financial markets. Yahoo Finance was selected as the data source because it offers accessible, consistent, and widely used financial data for academic and investment-related analysis.

Market	Company	Ticker	Industry
U.S.	Apple	AAPL	Consumer Technology
U.S.	Microsoft	MSFT	Software
U.S.	NVIDIA	NVDA	Semiconductors
Kazakhstan	Kaspi.kz	KSPI	FinTech
Kazakhstan	Halyk	HSBK.IL	Banking
Kazakhstan	KAP	KAP.IL	Uranium

The sample period covers five years, from January 1, 2020 to January 1, 2025. This period was selected because it includes different market conditions, including periods of high volatility, market recovery, technological growth, and changing macroeconomic environments. Using a five-year period allows the study to observe stock performance over a meaningful time horizon rather than focusing only on short-term price movements.

The selected U.S. stocks are Apple Inc. (AAPL), Microsoft Corporation (MSFT), and NVIDIA Corporation (NVDA). These companies were chosen because they are among the largest and most influential technology companies in the United States equity market. They also represent important areas of the modern economy, including consumer technology, software, cloud computing, semiconductors, and artificial intelligence.

The selected Kazakhstan-related stocks are Kaspi.kz (KSPI), Halyk Bank (HSBK.IL), and Kazatomprom (KAP.IL). These companies were selected because they represent major industries in Kazakhstan, including financial technology, banking, and natural resources. They are also among the most prominent firms connected to the Kazakhstan equity market. Therefore, they provide suitable examples for studying investment characteristics in an emerging-market context.

Overall, the selected companies represent major industries and are among the most prominent firms in their respective markets. This selection allows the study to compare risk, return, diversification, and trend-following strategy performance between a developed market and an emerging market.

3.2 Daily Return Calculation

The first step in the analysis is calculating daily stock returns. Stock prices alone show the absolute value of an asset, but returns provide a standardized measure of performance. Daily returns make it possible to compare stocks with different price levels because they measure percentage changes rather than absolute price changes.

The daily return is calculated using the following formula:

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

P_{t-1} represents the daily return at time t ,

P_t represents the closing price on the current trading day, and

P_{t-1} represents the closing price on the previous trading day.

Daily returns were calculated to measure the percentage change in stock prices between consecutive trading days. This method allows the study to evaluate how much each stock gained or lost each day. After calculating daily returns for each stock, the return series were used for further analysis, including mean return, volatility, correlation, portfolio construction, and moving-average strategy testing.

3.3 Risk Measurement

Risk in this study is measured using volatility. In financial analysis, volatility refers to the degree of variation in asset returns over time. A stock with high volatility experiences larger price movements, while a stock with low volatility has more stable returns. Volatility is important because investors are concerned not only with how much return they can earn, but also with how uncertain or unstable those returns are.

Volatility was measured using the standard deviation of daily returns. The standard deviation shows how far daily returns typically move away from their average value. A higher standard deviation indicates greater uncertainty and higher risk, while a lower standard deviation indicates more stable performance.

The standard deviation formula can be written as:

$$\sigma = \sqrt{\frac{\sum_{i=1}^n (R_i - \bar{R})^2}{n - 1}}$$

Where:

σ

represents volatility,

R_t

represents the daily return at time t ,

$\bar{R}$

represents the average daily return, and

n

represents the number of observations.

In this study, volatility is used to compare the risk levels of U.S. and Kazakhstan stocks. It also provides the foundation for calculating risk-adjusted return.

3.4 Risk-Adjusted Return

Risk-adjusted return is used to evaluate how much return an investment generates relative to the amount of risk taken. Looking only at average return may be misleading because a stock with a high return may also have very high volatility. Therefore, risk-adjusted return provides a more balanced measure of investment performance.

In this study, risk-adjusted return is calculated using the following formula:

$$\text{Risk-Adjusted Return} = \frac{\text{Mean Return}}{\text{Volatility}}$$

This metric measures how much return investors receive for each unit of risk. A higher value indicates that the stock or portfolio provides more return relative to its volatility. A lower value indicates that the return may not be sufficient compared with the level of risk.

This measure is applied to individual stocks, portfolios, and moving-average strategies. It allows the study to compare investment performance across different markets and strategies in a consistent way.

3.5 Correlation Analysis

Correlation analysis is used to examine the relationship between the returns of different stocks. In portfolio management, correlation is important because it helps determine whether diversification can reduce risk. If stocks move in the same direction, diversification benefits may be limited. If stocks move differently, combining them in a portfolio may reduce overall risk.

This study uses Pearson correlation coefficients to evaluate the relationships among stocks. Pearson correlation measures the strength and direction of the linear relationship between two return series.

The value of the correlation coefficient ranges from -1 to +1. A correlation close to +1 means that two stocks tend to move in the same direction. A correlation close to 0 means that the relationship between the two stocks is weak. A correlation close to -1 means that the two stocks tend to move in opposite directions.

Correlation matrices were created separately for the selected U.S. stocks and Kazakhstan stocks. This allows the study to compare whether diversification opportunities differ between the two markets.

3.6 Portfolio Construction

To evaluate diversification benefits, equal-weighted portfolios were constructed for both markets. An equal-weighted portfolio gives the same weight to each stock in the portfolio. This approach is simple, transparent, and suitable for comparing the basic effects of diversification.

For the U.S. portfolio, Apple, Microsoft, and NVIDIA were each assigned an equal weight of approximately 33.3%. For the Kazakhstan portfolio, Kaspi.kz, Halyk Bank, and Kazatomprom were also each assigned an equal weight of approximately 33.3%.

The portfolio return was calculated as the weighted average of the individual stock returns:

$$R_p = w_1R_1 + w_2R_2 + w_3R_3$$

Where:

$$R_p$$

represents the portfolio return,

$$w_i$$

represents the weight of each stock, and

$$R_i$$

represents the return of each stock.

Equal-weighted portfolios were constructed to evaluate diversification benefits. After constructing the portfolios, the study calculated portfolio mean return, volatility, and risk-adjusted return. This makes it possible to compare whether combining stocks improves investment performance relative to holding individual stocks.

3.7 Moving Average Strategies

Moving-average strategies were used to evaluate the effectiveness of trend-following approaches in both markets. A moving average smooths historical price data by calculating the average price over a specific number of trading days. It helps identify the general direction of price movement by reducing short-term noise.

This study applies four moving-average lengths: MA20, MA50, MA100, and MA200. MA20 represents a short-term moving average, MA50 and MA100 represent medium-term trends, and MA200 represents a long-term trend indicator. Comparing these different moving-average lengths allows the study to examine whether shorter or longer trend-following strategies perform better.

The trading rule used in this study is simple:

$$P_t > MA_n \Rightarrow Buy$$

$$P_t < MA_n \Rightarrow Sell$$

When the stock price is above its moving average, the strategy assumes that the asset is in an upward trend and holds the stock. When the stock price is below its moving average, the strategy assumes that the trend is weak or negative and exits the position.

Strategy returns were calculated by multiplying the trading signal by the daily return. To avoid using information from the same day, the signal was shifted by one trading day. This means that the strategy uses yesterday's signal to determine today's position. This approach makes the strategy

more realistic because investors cannot trade based on information that is only known after the trading day ends.

The cumulative performance of each moving-average strategy was then compared with a buy-and-hold approach. The study also calculated mean return, volatility, and risk-adjusted return for each strategy. This allows the research to evaluate whether moving-average strategies improve performance compared with simply holding the stock over the full period.

3.8 Data Analysis Tools

All analyses were conducted using Python in Jupyter Notebook. Python was selected because it is widely used in financial data analysis and provides powerful libraries for data collection, processing, calculation, and visualization.

The yfinance library was used to download historical stock price data from Yahoo Finance. Pandas was used for organizing the data, calculating daily returns, constructing portfolios, and creating correlation matrices. NumPy was used for numerical calculations, including volatility and risk-adjusted return calculations. Matplotlib was used to generate visualizations of stock prices, daily returns, and moving-average strategy performance.

Jupyter Notebook was used during the research process because it allows code, output, tables, and graphs to be displayed in one interactive environment. This made it suitable for exploring the data, testing formulas, and developing the quantitative analysis step by step.

Overall, the methodology follows a clear quantitative process. First, historical stock price data were collected. Second, daily returns were calculated. Third, risk and risk-adjusted performance were measured. Fourth, correlations were analyzed to evaluate diversification potential. Fifth, equal-weighted portfolios were constructed. Finally, moving-average strategies were tested to examine whether trend-following techniques are equally effective in developed and emerging equity markets.

Chapter 4: U.S. Market Analysis

4.1 Descriptive Statistics of U.S. Stocks

This section presents the descriptive statistics of the selected U.S. stocks: Apple, Microsoft, and NVIDIA. The analysis focuses on three key indicators: mean return, volatility, and risk-adjusted return. Mean return measures the average daily performance of each stock, volatility measures the risk or variability of returns, and risk-adjusted return shows how much return is generated for each unit of risk.

Table 4.1: Descriptive Statistics of U.S. Stocks

Stock	Mean Return	Volatility	Risk-Adjusted Return
Apple	0.001182	0.019956	0.059237
Microsoft	0.000988	0.019211	0.051418
NVIDIA	0.003050	0.033940	0.089859

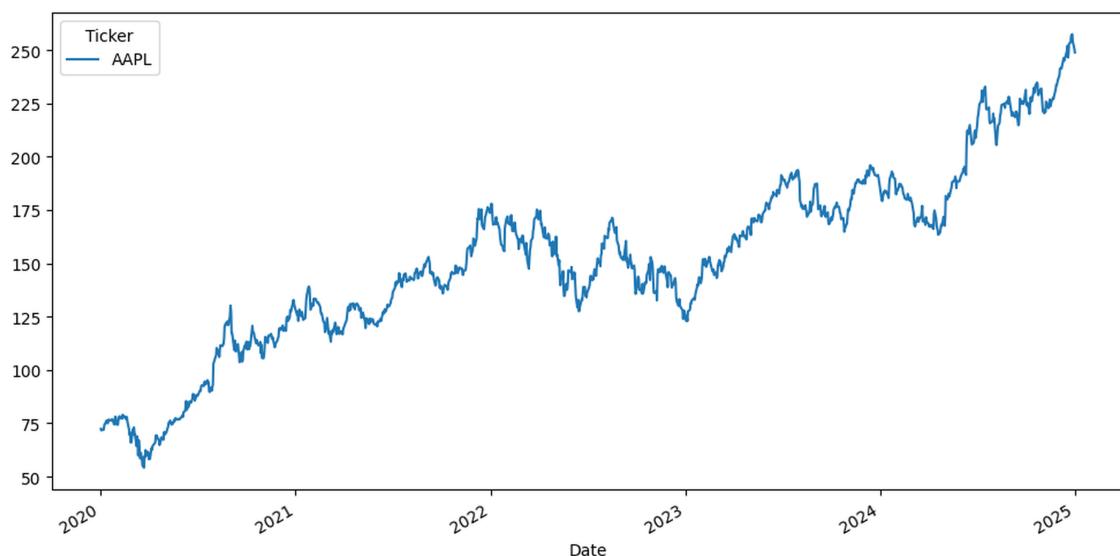
The results show that NVIDIA had the highest mean return among the three U.S. stocks. This indicates that NVIDIA generated the strongest average daily performance during the sample period. However, NVIDIA also had the highest volatility, meaning that its returns fluctuated more strongly than Apple and Microsoft.

Apple had the lowest volatility, suggesting that it was the least risky stock among the three based on daily return variation. Microsoft had the lowest mean return and the lowest risk-adjusted return. NVIDIA achieved the highest risk-adjusted return, which means that despite its high volatility, it provided the strongest return relative to risk.

4.2 Price Behavior

This section examines the price behavior of Apple, Microsoft, and NVIDIA from 2020 to 2024. Price charts were used to observe the general trend and long-term movement of each stock.

Figure 4.1: Apple Price Chart



Apple showed steady growth over the period from 2020 to 2024. Although the stock experienced short-term fluctuations, the overall trend was upward. This reflects Apple's strong market position, stable business performance, and continued investor confidence during the sample period.

Figure 4.2: Microsoft Price Chart

Microsoft also showed a strong upward trend during the sample period. The price movement suggests consistent growth, supported by Microsoft's expansion in software, cloud computing, and artificial intelligence-related business areas. Compared with Apple, Microsoft's trend was also stable, although daily price changes still occurred throughout the period.

Figure 4.3: NVIDIA Price Chart

NVIDIA experienced the most significant increase among the three U.S. stocks. Its price growth was especially strong in the later part of the sample period. This reflects the company's strong performance in semiconductors, graphics processing units, and artificial intelligence-related

demand. However, the strong increase was also accompanied by higher volatility, as shown in the descriptive statistics.

Overall, the price charts show that all three U.S. stocks increased over the sample period, but NVIDIA had the most dramatic growth.

4.3 Daily Return Analysis

Daily return analysis was conducted to understand the short-term fluctuations of each stock. Return plots and histograms were used to examine how daily returns were distributed over time.

Daily Return Plot of U.S. Stocks

The daily return plots show that most daily returns were relatively small, but occasional large positive and negative movements occurred. This pattern is common in equity markets, where most trading days have moderate price changes while extreme movements happen less frequently.

Return Histogram of U.S. Stocks

The return histograms show that most daily returns are concentrated around zero. This means that on most trading days, stock price changes were relatively small. Large price movements were relatively rare, but they still occurred during periods of market stress or strong positive momentum.

Among the three stocks, NVIDIA exhibited the widest return distribution. This supports the earlier finding that NVIDIA had the highest volatility. Apple and Microsoft had narrower distributions, indicating more stable daily return behavior compared with NVIDIA.

4.4 Correlation Analysis

Correlation analysis was used to examine the relationships among Apple, Microsoft, and NVIDIA returns. Pearson correlation coefficients were calculated to measure the strength of co-movement between the stocks.

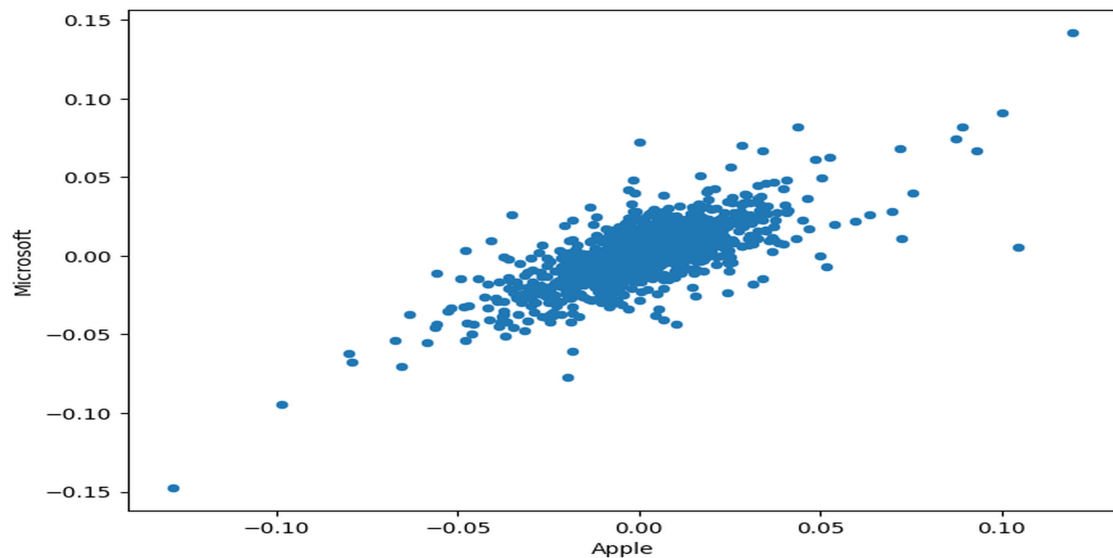
Table 4.2: Correlation Between U.S. Stocks

Pair	Correlation
Apple-Microsoft	0.748
Apple-NVIDIA	0.606
Microsoft-NVIDIA	0.682

The results show that Apple and Microsoft had the strongest correlation, with a coefficient of 0.748. This indicates that the two stocks often moved in the same direction. This may be because both companies are large technology firms and are affected by similar market factors, such as interest rates, investor sentiment, and expectations for technology-sector growth.

The correlation between Apple and NVIDIA was lower at 0.606, while the correlation between Microsoft and NVIDIA was 0.682. These values are still positive, but they are weaker than the Apple-Microsoft relationship. This suggests that NVIDIA provided additional diversification benefits within the U.S. stock portfolio.

Figure 4.4: Scatter Plot of Apple and Microsoft Returns



The scatter plot between Apple and Microsoft returns confirms their strong positive relationship. Many observations move in the same general direction, showing that the two stocks often experience similar return patterns.

4.5 Portfolio Analysis

An equal-weighted portfolio was constructed using Apple, Microsoft, and NVIDIA. Each stock was assigned a weight of approximately 33.3%. The purpose of this portfolio was to evaluate whether diversification improved risk-return performance compared with individual stocks.

Table 4.3: U.S. Equal-Weighted Portfolio Performance

Metric	Value
Mean Return	0.00174
Volatility	0.02161
Risk-adjusted Return	0.0805

The portfolio mean return was 0.00174, while the portfolio volatility was 0.02161. Compared with the individual stocks, the portfolio return was higher than Apple and Microsoft but lower than NVIDIA. However, the portfolio volatility was lower than NVIDIA's volatility.

This shows that diversification reduced risk compared with holding NVIDIA alone. Although the portfolio still had some exposure to high-growth technology stocks, combining Apple, Microsoft, and NVIDIA helped reduce overall volatility. Portfolio volatility increased less than returns, suggesting that diversification improved the balance between risk and return.

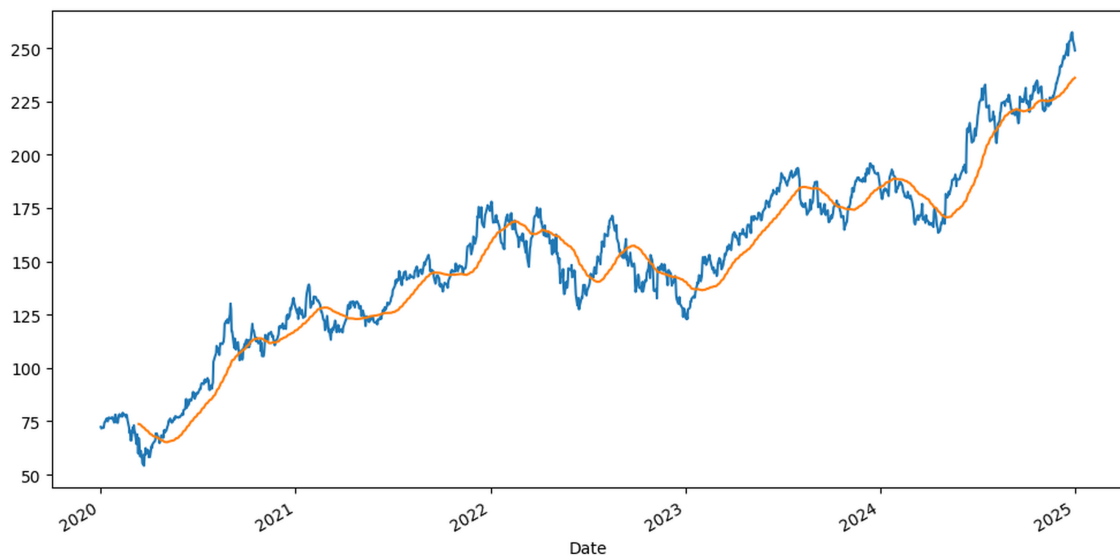
4.6 Moving Average Strategy

Moving-average strategies were tested to evaluate whether trend-following rules improved performance compared with a buy-and-hold approach. Four moving-average lengths were used: MA20, MA50, MA100, and MA200.

The trading rule was simple: if the stock price was above the moving average, the strategy generated a buy signal. If the stock price was below the moving average, the strategy exited the position. This approach was used to test whether following price trends could improve risk-adjusted performance.

Table 4.4: Apple Moving-Average Strategy Performance

Strategy	Mean Return	Volatility	Return/Risk
Buy and Hold	0.001182	0.019956	0.059230
MA20	0.001071	0.013094	0.081793
MA50	0.000737	0.013076	0.056363
MA100	0.000717	0.013689	0.052378
MA200	0.000491	0.012534	0.039173



For Apple, the MA20 strategy achieved the highest risk-adjusted return. Although its mean return was slightly lower than buy-and-hold, its volatility was much lower. This suggests that the MA20 strategy helped reduce risk and improved risk-adjusted performance. However, longer moving averages such as MA100 and MA200 produced lower returns and weaker return/risk ratios.

Table 4.5: Microsoft Moving-Average Strategy Performance

Strategy	Mean Return	Volatility	Return/Risk
Buy and Hold	0.000988	0.019211	0.051418
MA20	0.000201	0.012198	0.016480

MA50	0.000346	0.012351	0.027992
MA100	0.000473	0.012276	0.038540
MA200	0.000434	0.011348	0.038273

For Microsoft, all moving-average strategies had lower mean returns than buy-and-hold. Although the strategies reduced volatility, the reduction in return was larger. As a result, none of the moving-average strategies outperformed buy-and-hold in terms of risk-adjusted return.

Table 4.6: NVIDIA Moving-Average Strategy Performance

Strategy	Mean Return	Volatility	Return/Risk
Buy and Hold	0.003050	0.033940	0.089859
MA20	0.002216	0.024446	0.090669
MA50	0.002122	0.025424	0.083451
MA100	0.002141	0.025230	0.084855
MA200	0.002190	0.025558	0.085703

For NVIDIA, the MA20 strategy produced a risk-adjusted return that was slightly higher than buy-and-hold. This suggests that short-term trend-following helped reduce volatility while maintaining relatively strong returns. However, MA50, MA100, and MA200 did not outperform buy-and-hold based on return/risk.

Overall, the moving-average results show that strategy performance differed across stocks. MA strategies reduced volatility in many cases, but they did not consistently outperform buy-and-hold. Shorter moving averages, especially MA20, performed better for Apple and NVIDIA, while Microsoft's buy-and-hold strategy remained stronger.

4.7 Chapter Summary

This chapter presented the empirical results for the selected U.S. stocks. The analysis showed that NVIDIA exhibited the highest return and highest volatility, while Apple had the lowest volatility. NVIDIA also achieved the highest risk-adjusted return among the three stocks.

The price analysis showed that Apple and Microsoft experienced steady upward trends, while NVIDIA had the most significant increase. Daily return analysis showed that most returns were concentrated around zero, and large price movements were relatively rare. NVIDIA had the widest return distribution, confirming its higher volatility.

Correlation analysis showed that Apple and Microsoft had the strongest relationship, while NVIDIA provided additional diversification benefits. The equal-weighted portfolio reduced risk compared with holding NVIDIA alone and improved the balance between return and volatility.

Finally, the moving-average strategy analysis showed that performance differed across stocks and moving-average lengths. MA strategies reduced volatility in several cases, but they did not consistently outperform buy-and-hold. Therefore, the results suggest that moving-average strategies

may be useful in some cases, but their effectiveness depends on the stock and the chosen moving-average length.

Chapter 5: Kazakhstan Market Analysis

5.1 Descriptive Statistics of Kazakhstan Stocks

This chapter presents the empirical results for the selected Kazakhstan-related stocks: Kaspi.kz, Halyk Bank, and Kazatomprom. The analysis uses mean return, volatility, and risk-adjusted return to evaluate the performance of each stock.

Table 5.1: Descriptive Statistics of Kazakhstan Stocks

Stock	Mean Return	Volatility	Return/Risk
Kaspi	0.000526	0.0251	0.0210
Halyk	0.001232	0.0272	0.0453
KAP	0.001508	0.0288	0.0524

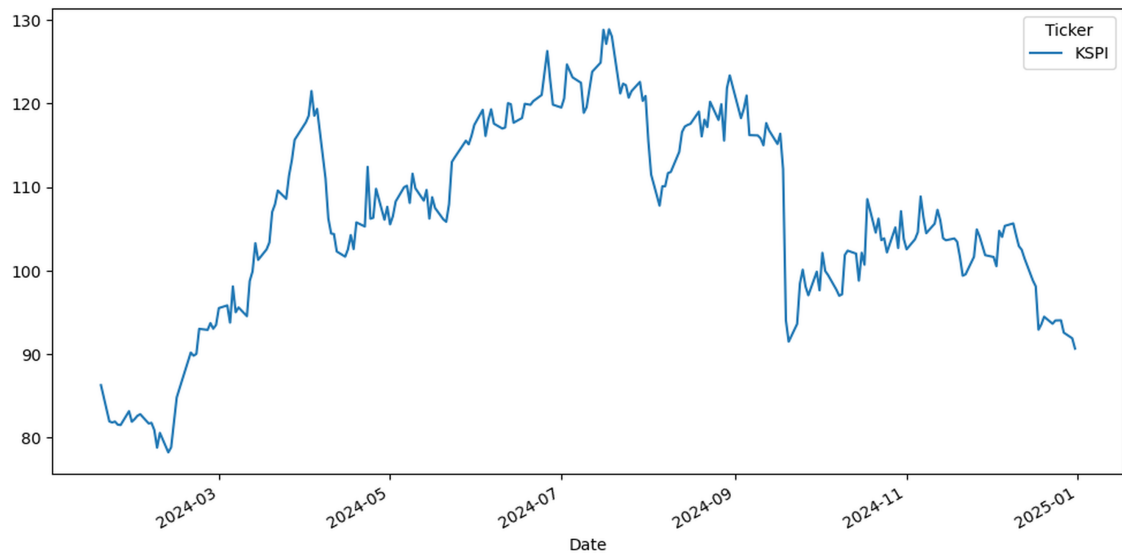
The results show that KAP achieved the highest average return among the three Kazakhstan stocks. This suggests that Kazatomprom provided the strongest return performance during the sample period. However, KAP also had the highest volatility, indicating that its higher return was associated with greater risk.

Kaspi exhibited the lowest risk-adjusted performance, with a return/risk ratio of 0.0210. Although Kaspi is one of the most prominent companies connected to Kazakhstan's equity market, its average return was relatively low compared with its level of volatility.

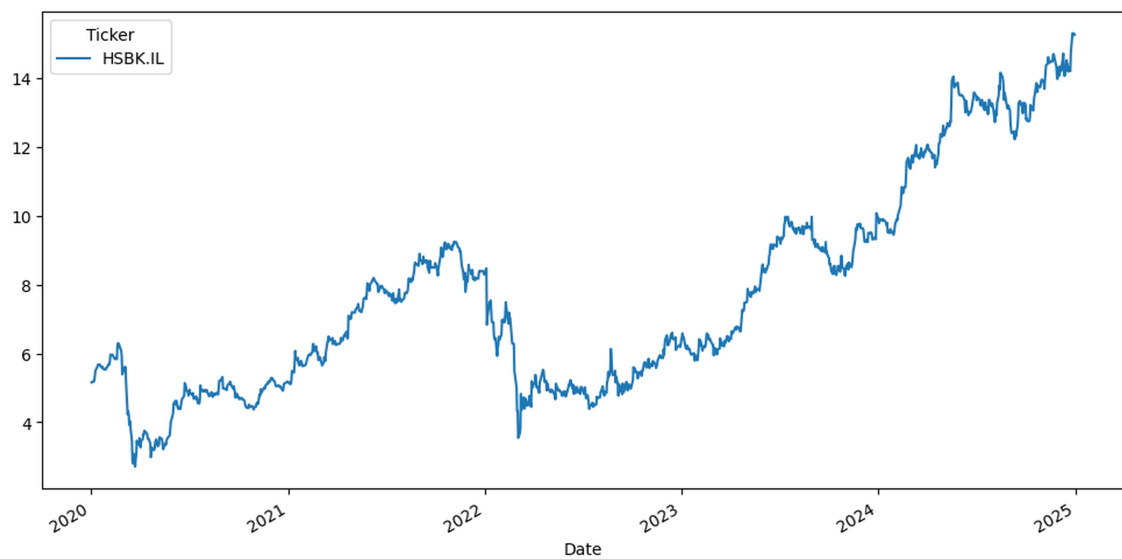
Halyk demonstrated a balance between risk and return. Its mean return was higher than Kaspi's, while its volatility was slightly lower than KAP's. As a result, Halyk produced a moderate risk-adjusted return compared with the other two stocks.

5.2 Price Behavior

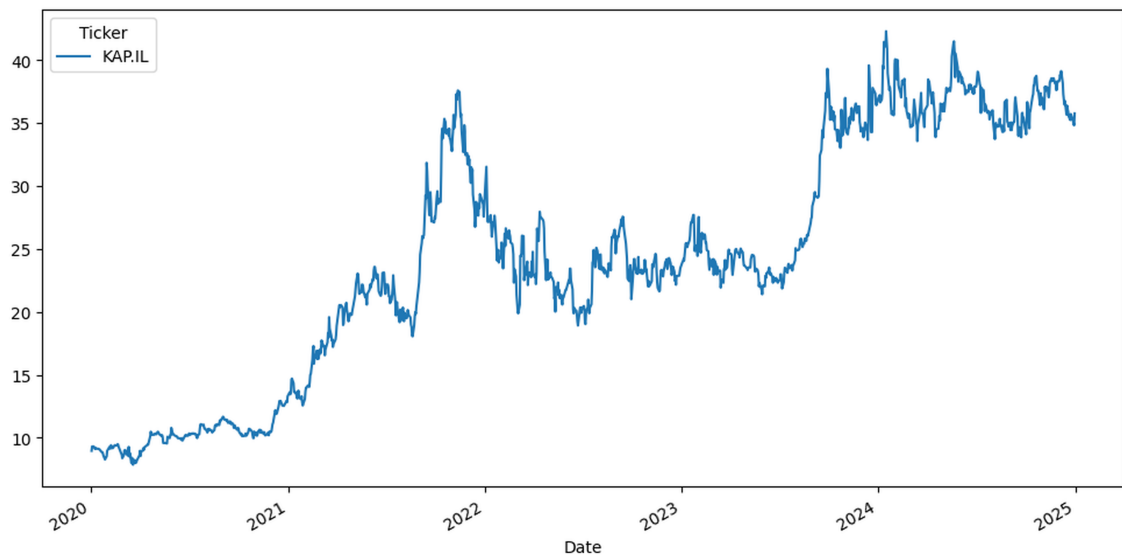
The price behavior of the three Kazakhstan stocks shows different patterns across the sample period.



Kaspi experienced a noticeable increase between March 2024 and July 2024. However, after July 2024, the stock price declined. This suggests that Kaspi's price movement was not stable throughout the period and may have been affected by changing investor expectations or market conditions.



Halyk experienced a sharp decline during the COVID-19 crisis in 2020. This reflects the strong negative impact of the pandemic on financial markets. However, from 2022 onward, Halyk showed a strong upward movement, indicating improved investor confidence and stronger market performance.



KAP showed a sharp increase in 2021, followed by a significant decline in 2022. After 2023, the stock continued to show noticeable volatility. This pattern suggests that KAP was strongly affected by market cycles and sector-specific factors, particularly those related to natural resources and uranium demand.

Overall, the price charts indicate that Kazakhstan stocks experienced greater price volatility and less stable long-term growth than large U.S. technology stocks. Unlike large U.S. technology stocks, their price behavior was more irregular and more sensitive to specific market events.

5.3 Daily Return Analysis

Daily return plots were used to examine the short-term volatility of Kaspi, Halyk, and KAP.

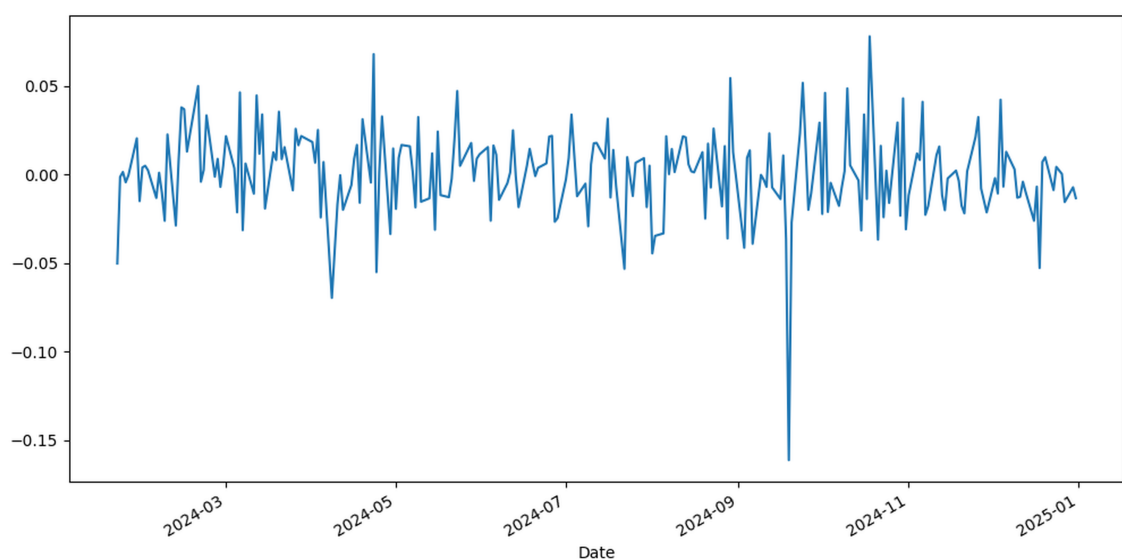


Figure 5.1: Kaspi Daily Return Plot

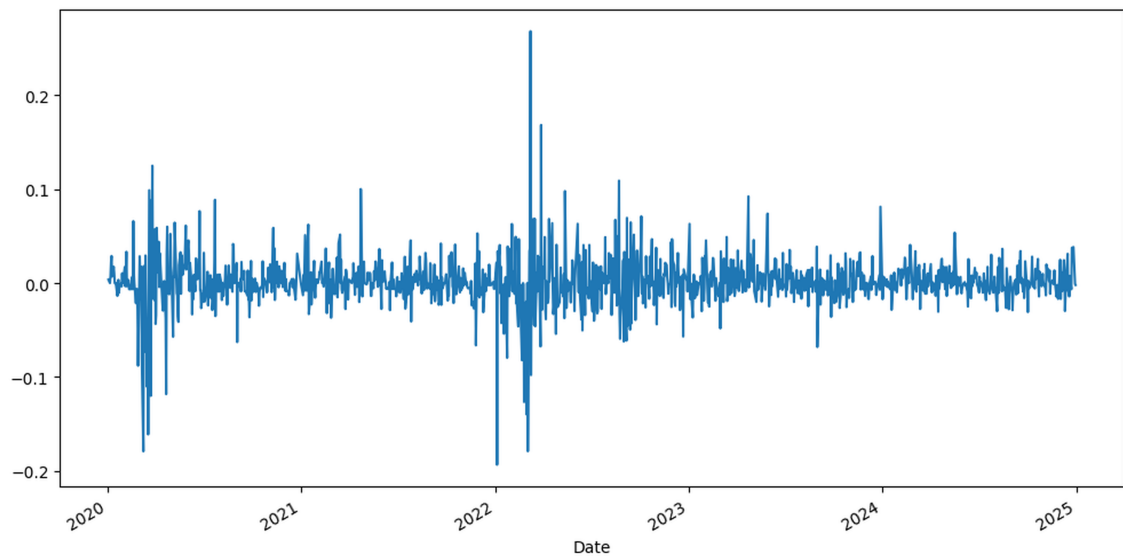


Figure 5.2: Halyk Daily Return Plot

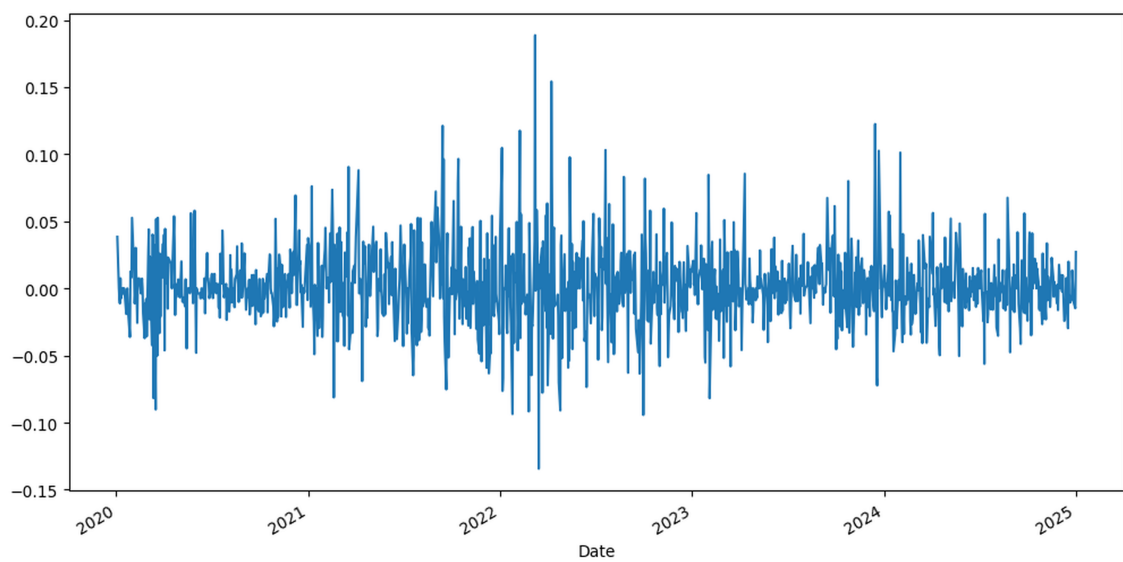


Figure 5.3: KAP Daily Return Plot

All three stocks exhibited considerable volatility during the sample period. Their daily returns frequently moved around zero, but several large positive and negative return movements were also observed.

Halyk and KAP experienced the largest fluctuations during 2022. This suggests that 2022 was a particularly volatile period for Kazakhstan-related equities. The fluctuations may reflect global uncertainty, commodity market changes, inflation pressure, geopolitical development in Eastern Europe and global commodity market uncertainty, and changing investor sentiment toward emerging markets.

Kaspi's returns were concentrated around zero but showed several extreme movements. This means that while most daily price changes were relatively small, Kaspi also experienced occasional large shocks. These return patterns confirm that Kazakhstan stocks carried meaningful short-term risk.

5.4 Correlation Analysis

Correlation analysis was conducted to examine the relationships among the selected Kazakhstan stocks. Pearson correlation coefficients were used to measure how strongly the stocks moved together.

Table 5.2: Correlation Between Kazakhstan Stocks

Pair	Correlation
Kaspi-Halyk	0.100
Kaspi-KAP	0.086
Halyk-KAP	0.184

The results show that Kazakhstan stocks had very low correlations with each other. The correlation between Kaspi and KAP was only 0.086, while the correlation between Kaspi and Halyk was 0.100. The highest correlation was between Halyk and KAP, at 0.184, but this value was still relatively low.

This is one of the most important findings of the Kazakhstan market analysis. Compared with the U.S. stocks, whose correlations were between approximately 0.6 and 0.75, the Kazakhstan stocks exhibited substantially lower correlations. This suggests greater diversification potential in the Kazakhstan portfolio.

Low correlation means that the stocks did not move closely together. Therefore, combining these stocks in a portfolio may reduce overall risk more effectively than combining highly correlated stocks. This result supports the idea that diversification benefits may be stronger in the Kazakhstan sample than in the U.S. technology stock sample.

5.5 Portfolio Analysis

An equal-weighted Kazakhstan portfolio was constructed using Kaspi, Halyk, and KAP. Each stock was assigned an equal weight of approximately 33.3%.

Table 5.3: Kazakhstan Equal-Weighted Portfolio Performance

Metric	Value
Mean Return	0.000792
Volatility	0.01322
Return/Risk	0.05989

The equal-weighted Kazakhstan portfolio achieved a mean return of 0.000792 and volatility of 0.01322. The portfolio volatility was lower than the volatility of each individual stock. This demonstrates clear diversification benefits.

The lower portfolio volatility can be explained by the weak correlations among the three stocks. Since Kaspi, Halyk, and KAP did not move closely together, combining them reduced overall portfolio risk. Although the portfolio mean return was lower than the individual returns of Halyk and KAP, the reduction in volatility improved the overall risk-return balance.

Therefore, the Kazakhstan portfolio provides strong evidence that diversification can reduce risk when assets have low correlations.

5.6 Moving Average Strategies

Moving-average strategies were tested for Kazakhstan stocks to evaluate whether trend-following rules improved performance. The strategies were based on moving averages such as MA20, MA50, and MA100, depending on the available results for each stock.

Table 5.4: Kaspi Moving-Average Strategy Performance

Strategy	Return/Risk
Buy and Hold	0.021
MA20	0.021
MA50	-0.089
MA100	-0.026

For Kaspi, moving-average strategies did not outperform buy-and-hold. The MA20 strategy produced a similar return/risk ratio to buy-and-hold, while MA50 and MA100 produced negative results. This suggests that trend-following strategies were not effective for Kaspi during the sample period.

Table 5.5: Halyk Moving-Average Strategy Performance

Strategy	Return/Risk
Buy and Hold	0.045
MA20	0.045
MA50	0.056

For Halyk, the MA50 strategy slightly outperformed buy-and-hold. The MA20 strategy produced almost the same return/risk ratio as buy-and-hold, while MA50 provided a modest improvement. This suggests that a medium-term moving average was more effective for Halyk than a shorter-term moving average.

Table 5.6: KAP Moving-Average Strategy Performance

Strategy	Return/Risk
Buy and Hold	0.052
MA20	0.008
MA50	0.026

For KAP, trend-following strategies underperformed significantly. Both MA20 and MA50 produced lower return/risk ratios than buy-and-hold. This indicates that moving-average strategies failed to capture KAP's price movements effectively during the sample period.

Overall, the moving-average results for Kazakhstan stocks were mixed. In most cases, buy-and-hold performed better than trend-following strategies. Only Halyk's MA50 strategy showed slight improvement over buy-and-hold. These findings suggest that moving-average strategies may not consistently improve performance in Kazakhstan equities. These findings suggest that trend-following strategies may be less effective in relatively inefficient or highly volatile emerging equity markets.

5.7 Chapter Summary

This chapter analyzed the selected Kazakhstan stocks: Kaspi, Halyk, and KAP. The descriptive statistics showed that KAP generated the highest average return, while Kaspi had the lowest risk-adjusted performance. Halyk showed a more balanced relationship between risk and return.

The price analysis showed that Kazakhstan stocks experienced significant fluctuations. Kaspi increased between March and July 2024 before declining, Halyk recovered strongly after the COVID-19 period, and KAP showed sharp movements in 2021 and 2022.

The correlation analysis produced one of the most important findings of this chapter. Kazakhstan stocks exhibited much lower correlations than U.S. equities, suggesting stronger diversification potential. The equal-weighted Kazakhstan portfolio achieved lower volatility than the individual stocks, confirming the benefits of diversification.

Finally, moving-average strategies produced mixed results. Halyk's MA50 strategy slightly outperformed buy-and-hold, but Kaspi and KAP did not show consistent improvement from trend-following strategies. Overall, buy-and-hold generally outperformed moving-average strategies in the Kazakhstan sample.

Chapter 6: Comparative Analysis and Discussion

6.1 Risk and Return Comparison

This section compares the risk and return characteristics of the selected U.S. and Kazakhstan stocks. The purpose is to evaluate whether the developed U.S. market and the emerging Kazakhstan market show different performance patterns.

Table 6.1: Risk and Return Comparison Between U.S. and Kazakhstan Stocks

Metric	U.S.	Kazakhstan
Highest Mean Return	NVIDIA (0.00305)	KAP (0.00151)
Lowest Volatility	Apple	Kaspi
Highest Return/Risk	NVIDIA	KAP

The results show that U.S. stocks generally generated higher returns than Kazakhstan stocks. NVIDIA achieved the highest average return among all selected assets, with a mean daily return of 0.00305. This was higher than KAP, which had the highest mean return among the Kazakhstan stocks at 0.00151.

In terms of volatility, Apple had the lowest volatility among the U.S. stocks, while Kaspi had the lowest volatility among the Kazakhstan stocks. This indicates that both markets contained relatively more stable assets, although the overall behavior of each market was different.

The highest risk-adjusted return in the U.S. sample was achieved by NVIDIA, while KAP achieved the highest risk-adjusted return in the Kazakhstan sample. Although U.S. technology stocks produced stronger returns, Kazakhstan stocks also provided attractive risk-adjusted performance despite operating in a smaller and less developed market.

Overall, the U.S. market offered stronger return potential, especially through NVIDIA. However, the Kazakhstan market also showed meaningful investment opportunities, particularly through KAP and Halyk.

6.2 Correlation and Diversification

Correlation analysis is one of the most important parts of this study because it helps explain diversification benefits. Assets with high correlations tend to move together, which limits diversification. Assets with low correlations move more independently, which can reduce portfolio risk when combined.

Table 6.2: Correlation Range Comparison

Market	Correlation Range
U.S.	0.606-0.748
Kazakhstan	0.086-0.184

The Kazakhstan stocks exhibited substantially lower correlations than the U.S. technology stocks. The U.S. stock correlations ranged from 0.606 to 0.748, showing that Apple, Microsoft, and

NVIDIA tended to move in the same direction. This is expected because all three firms belong to the technology sector and are influenced by similar market factors, including interest rates, investor sentiment, innovation cycles, and expectations for artificial intelligence and cloud computing growth.

In contrast, the Kazakhstan stock correlations ranged from only 0.086 to 0.184. These values indicate weak relationships among Kaspi, Halyk, and KAP. The low correlations suggest that the Kazakhstan stocks were less likely to move together. As a result, combining them in a portfolio reduced overall portfolio risk more effectively.

This is one of the strongest findings of the study. Although Kazakhstan is a smaller and less developed market, the selected Kazakhstan stocks provided stronger diversification potential than the selected U.S. technology stocks. This shows that emerging markets may offer important diversification benefits, especially when stocks come from different sectors such as fintech, banking, and natural resources.

6.3 Portfolio Performance

Portfolio analysis was conducted to compare how diversification affected the U.S. and Kazakhstan samples. Equal-weighted portfolios were constructed for both markets.

Table 6.3: Portfolio Performance Comparison

Metric	U.S. Portfolio	Kazakhstan Portfolio
Mean Return	0.00174	0.00079
Volatility	0.02161	0.01322
Return/Risk	0.0805	0.0599

The U.S. portfolio generated higher average returns than the Kazakhstan portfolio. This result is consistent with the strong performance of U.S. technology stocks, especially NVIDIA. The U.S. portfolio mean return was 0.00174, compared with 0.00079 for the Kazakhstan portfolio.

However, the Kazakhstan portfolio exhibited lower volatility. Its standard deviation was 0.01322, compared with 0.02161 for the U.S. portfolio. This means that the Kazakhstan portfolio had more stable daily returns during the sample period.

Both portfolios benefited from diversification, but the Kazakhstan portfolio achieved greater risk reduction due to weaker correlations among its constituent stocks. The U.S. portfolio still reduced risk compared with holding NVIDIA alone, but the diversification effect was limited by the relatively high correlations among Apple, Microsoft, and NVIDIA.

The return/risk ratio was higher for the U.S. portfolio, showing stronger risk-adjusted performance overall. Nevertheless, the Kazakhstan portfolio demonstrated that lower correlation can produce meaningful risk reduction, even when individual stocks are volatile.

6.4 Moving Average Strategy Comparison

Moving-average strategies were tested to evaluate whether trend-following approaches improved investment performance in both markets. The strategies used simple rules based on MA20, MA50, MA100, and MA200 where applicable.

Table 6.4: Best Moving-Average or Investment Strategy by Stock

Stock	Best Strategy
Apple	MA20
Microsoft	MA50/MA100
NVIDIA	Buy and Hold
Kaspi	Buy and Hold
Halyk	MA50
KAP	Buy and Hold

The results show that moving-average strategies produced mixed results in both markets. For Apple, the MA20 strategy improved risk-adjusted performance by reducing volatility. For Microsoft, medium-term strategies such as MA50 or MA100 performed relatively better than shorter strategies, although buy-and-hold remained competitive. For NVIDIA, buy-and-hold was stronger because the stock experienced substantial long-term growth.

In the Kazakhstan sample, Kaspi and KAP did not benefit meaningfully from moving-average strategies. Buy-and-hold generally performed better for these stocks. Halyk was the main exception, as the MA50 strategy slightly outperformed buy-and-hold.

These findings suggest that moving-average strategies do not consistently outperform passive investment approaches. Their effectiveness depends on the stock, the market environment, and the selected moving-average length. Trend-following may reduce downside risk in some cases, but it can also miss large upward price movements.

6.5 Discussion

The first major finding is that developed-market stocks generated higher returns. The U.S. technology stocks, especially NVIDIA, produced stronger average returns than the Kazakhstan stocks. This reflects the strong growth of the U.S. technology sector during the sample period.

The second major finding is that emerging-market stocks provided stronger diversification benefits. Although Kazakhstan stocks did not generate the same level of average return as U.S. technology stocks, their correlations were much lower. This allowed the Kazakhstan portfolio to achieve lower volatility than the individual stocks.

The third finding is that trend-following strategies were not universally effective. Moving-average strategies worked better for some stocks, such as Apple and Halyk, but they did not consistently outperform buy-and-hold. This suggests that simple technical trading rules should be used carefully and should not be assumed to produce better results in all markets.

The fourth finding is that market structure influences investment performance. The U.S. sample consisted of large technology companies that were strongly affected by similar global growth themes. This helped explain their higher correlations. The Kazakhstan sample included companies from different sectors, including fintech, banking, and natural resources. This sector difference contributed to lower correlations and stronger diversification benefits.

Overall, the comparison shows that developed and emerging markets offer different investment advantages. The U.S. market provided stronger growth and higher returns, while the Kazakhstan market provided stronger diversification potential.

These findings suggest that investors may benefit from considering both developed and emerging markets when constructing diversified portfolios, as each market offers distinct advantages.

6.6 Limitations

This study has several limitations. First, only six stocks were examined: three from the United States and three from Kazakhstan. A larger sample could provide a more complete picture of each market.

Second, the sample period was limited to 2020-2025. Different results may be obtained if a longer time period is used or if the analysis covers different market cycles.

Third, transaction costs were not considered. In real trading, commissions, bid-ask spreads, taxes, and liquidity constraints may reduce the profitability of moving-average strategies.

Fourth, the moving-average strategies were limited to simple rules. More advanced strategies could include stop-loss rules, position sizing, transaction cost adjustments, or combinations of technical indicators.

Because of these limitations, the results should be interpreted as evidence from a focused sample rather than as a complete conclusion about all U.S. and Kazakhstan equities.

6.7 Summary

This chapter compared the U.S. and Kazakhstan equity markets. The results showed that U.S. stocks generally generated higher returns, while Kazakhstan stocks provided stronger diversification benefits due to lower correlations. Portfolio analysis demonstrated the importance of diversification in both markets. The U.S. portfolio achieved higher returns and stronger risk-adjusted performance, while the Kazakhstan portfolio achieved lower volatility.

Moving-average strategies produced mixed results. Some stocks benefited from trend-following rules, but buy-and-hold generally remained competitive and often performed better. These findings highlight the differences between developed and emerging equity markets and provide insight into how market characteristics influence investment performance.

Chapter 7: Conclusion

7.1 Research Objectives

The primary objective of this study was to conduct a comparative quantitative analysis of selected U.S. and Kazakhstan stocks. Specifically, the study examined differences in risk and return characteristics, diversification benefits, portfolio performance, and the effectiveness of moving-average-based trend-following strategies across developed and emerging equity markets.

To achieve this objective, three U.S. technology stocks (Apple, Microsoft, and NVIDIA) and three Kazakhstan-related stocks (Kaspi.kz, Halyk Bank, and Kazatomprom) were selected. Daily stock price data from 2020 to 2025 were analyzed using descriptive statistics, correlation analysis, portfolio construction, and moving-average strategies. Through this analysis, the study aimed to identify how investment characteristics differ between the two market environments and whether investors may benefit from combining assets from developed and emerging markets.

7.2 Overall Conclusions

Overall, the findings of this study indicate that developed and emerging markets provide different investment advantages.

The U.S. market demonstrated stronger return potential, primarily due to the exceptional performance of large technology companies such as NVIDIA. The U.S. portfolio generated a higher average daily return (0.00174) than the Kazakhstan portfolio (0.00079), while the Kazakhstan portfolio exhibited a lower standard deviation (0.0132 versus 0.0216), suggesting greater portfolio stability and diversification advantages. Furthermore, the lower correlations observed among Kazakhstan stocks further strengthened these diversification effects.

Another important finding is that moving-average strategies did not consistently outperform buy-and-hold strategies. While selected stocks benefited from trend-following approaches, passive investment remained highly competitive across both markets.

Taken together, these results suggest that investors should not evaluate markets solely based on their level of development. Instead, factors such as correlation, volatility, and portfolio construction may be equally important when making investment decisions.

7.3 Limitations and Future Research

This study is subject to several limitations. Only six stocks were included in the analysis, and the sample period was limited to 2020–2025. In addition, transaction costs and liquidity constraints were not considered.

Future research may expand the stock sample, include additional emerging markets, and examine more advanced portfolio optimization and technical trading strategies. Future studies could employ machine-learning models, such as Long Short-Term Memory (LSTM) networks or Random Forest algorithms, to improve stock return forecasting and portfolio allocation.

7.4 Final Remarks

In conclusion, the results of this study suggest that in an increasingly interconnected global financial system, investors who understand the complementary characteristics of developed and emerging markets may be better positioned to construct resilient and efficient portfolios in the face of future uncertainty.

While developed markets offer stronger growth opportunities, emerging markets provide diversification benefits that may improve portfolio stability. Investors may therefore benefit from combining assets from both market environments to achieve a more balanced trade-off between risk and return.

As financial markets continue to evolve, understanding the differences among markets will remain an essential component of effective investment decision-making.

Consequently, the study successfully addressed all research questions concerning risk-return trade-offs, diversification benefits, and the effectiveness of moving-average strategies across different market environments.